

Schizophrenia and Other Psychotic Disorders / 2

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Causes of Schizophrenia:

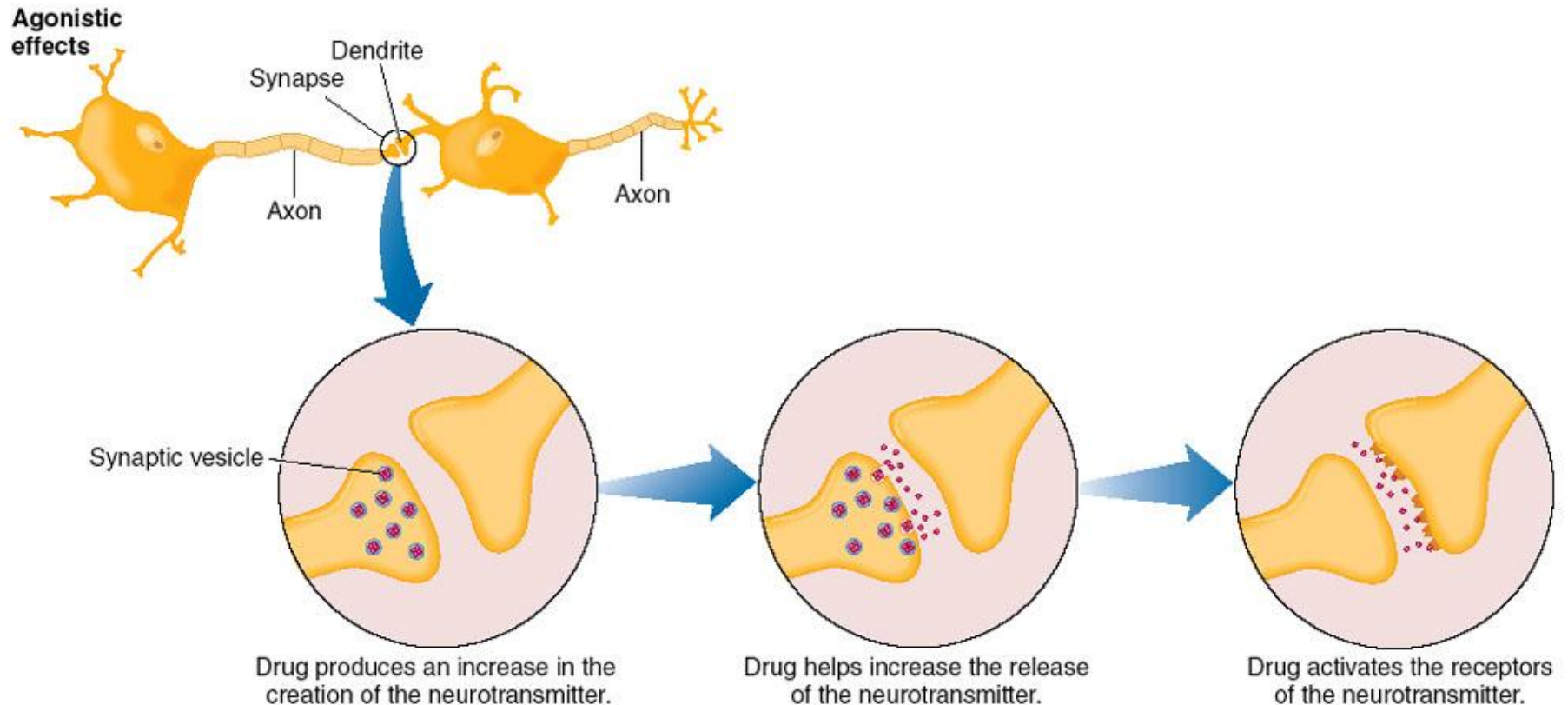
Findings From Genetic Research

- Family Studies
 - Inherit a tendency for schizophrenia
 - Do not inherit specific forms of schizophrenia
 - Risk increases with genetic relatedness
- Twin Studies
 - Monozygotic twins – Risk for schizophrenia is 48%
 - Fraternal (dizygotic) twins – Risk drops to 17%
 - Adoption Studies -- Risk for schizophrenia remains high
 - Cases where a biological parent has schizophrenia
- Summary of Genetic Research
 - Risk for schizophrenia increases with genetic relatedness
 - Risk is transmitted independently of diagnosis
 - Strong genetic component does not explain everything

Causes of Schizophrenia: Neurotransmitter Influences

- The Dopamine Hypothesis
 - Drugs that increase dopamine (agonists)
 - Result in schizophrenic-like behavior (L-Dopa for Parkinson's disease)
 - Drugs that decrease dopamine (antagonists)
 - Reduce schizophrenic-like behavior (Neuroleptics)
 - Current theories – Emphasize many neurotransmitters (Serotonin, GABA, & Glutamate) also have a role

Causes of Schizophrenia: Neurotransmitter Influences (cont.)



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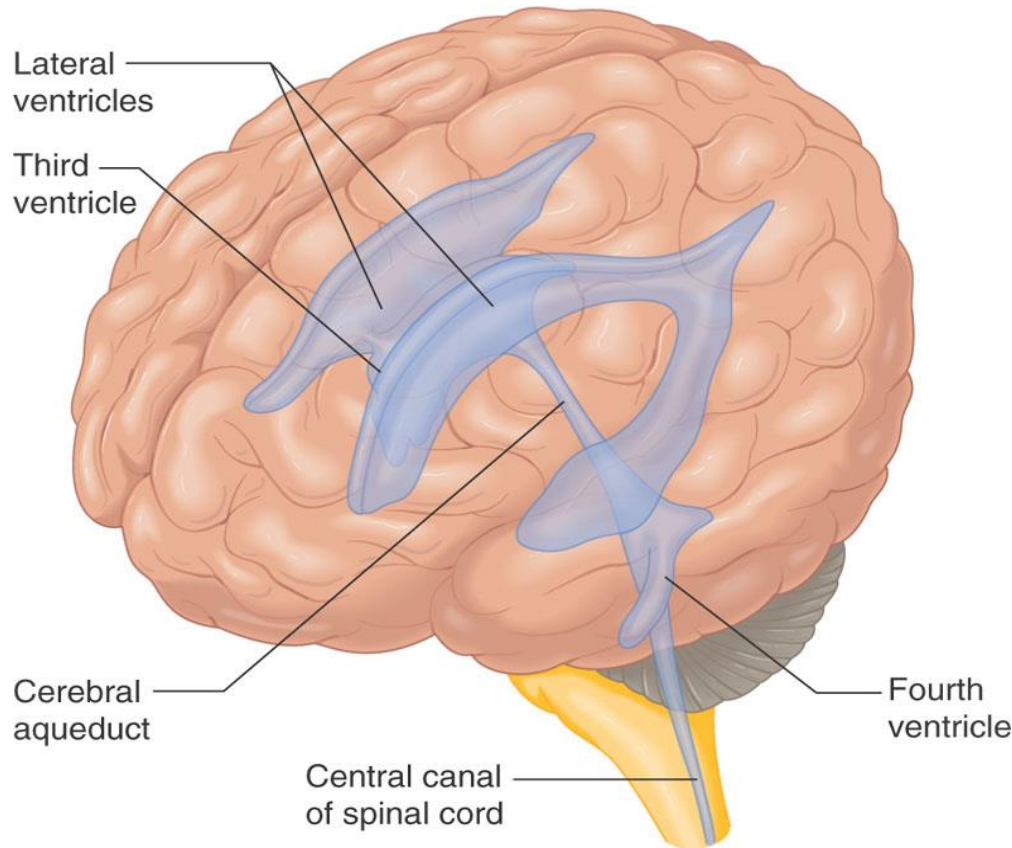
Other Neurobiological Influences

- Structural and Functional Abnormalities in the Brain
 - Enlarged ventricles and reduced tissue volume
 - Hypofrontality – Less active frontal lobes
 - A major dopamine pathway
- Viral Infections during early prenatal development
 - Findings are inconclusive

Structural and functional brain abnormalities

- Not unique to schizophrenia

Causes of Schizophrenia: Other Neurobiological Influences (cont.)



Causes of Schizophrenia: Psychological and Social Influences

- The Role of Stress
 - May activate underlying vulnerability
 - May also increase risk of relapse
- Family Interactions
 - Families – Show ineffective communication patterns
 - High expressed emotion – Associated with relapse
- The Role of Psychological Factors
 - Exert only a minimal effect in producing schizophrenia

Causes of Schizophrenia: Neurodevelopmental Model

- Neurodevelopmental model supposes in schizophrenia the presence of “silent lesion” in the brain, mostly in the parts, important for the development of integration (frontal, parietal and temporal), which is caused by different factors (genetic, inborn, infection, trauma...) during very early development of the brain in prenatal or early postnatal period of life.
- It does not interfere too much with the basic brain functioning in early years, but expresses itself in the time, when the subject is stressed by demands of growing needs for integration, during formative years in adolescence and young adulthood.

Treatment of Schizophrenia

- The acute schizophrenic patients will respond usually to antipsychotic medication
- Development of Antipsychotic (Neuroleptic) Medications
 - Often the first line treatment for schizophrenia
 - Began in the 1950s
 - Most reduce or eliminate positive symptoms
 - Acute and permanent side effects; (Extrapyramidal and Parkinson-like side effects, Tardive dyskinesia)
 - Compliance with medication is often a problem
- According to current consensus we use in the first line therapy the newer atypical antipsychotics, because their use is not complicated by appearance of extrapyramidal side-effects, or these are much lower than with classical antipsychotics.

- Conventional antipsychotics - (classical neuroleptics);

Chlorpromazine, Clopenthixole, Levopromazine, Thioridazine, Droperidole, Flupentixol, Fluphenazine, Haloperidol, Perphenazine, Pimozide, Prochlorperazine, Trifluoperazine,...

- Depot antipsychotics: (Fluphenazine deconate-Modecate), Flupentixol, and Zuclopenthixole
- Atypical antipsychotics - (new neuroleptics);

Amisulpiride, Clozapine, Olanzapine, Quetiapine, Risperidone, Sertindole, Sulpiride, Cariprazine,...

Psychosocial Treatment of Schizophrenia

- Psychosocial Approaches:
 - Behavioral (i.e., token economies) on inpatient units
 - Community care programs
 - Social and living skills training
 - Behavioral family therapy
 - Vocational rehabilitation

Electroconvulsive therapy (E.C.T) is also used in the treatment of schizophrenia, but may be useful when catatonia or prominent affective symptoms are present

Treating Schizophrenia

TREATMENT OF SCHIZOPHRENIA

Individual, Group, and Family Therapy

- Can help patient and family to understand the disease and symptom triggers.
- Teaches families communication skills.
- Provides resources for dealing with emotional and practical challenges.

Social Skills Training

- Can occur in hospital or community settings.
- Teaches the person with schizophrenia social, self-care, and vocational skills.

Medications

- Taking *neuroleptic medications* may help people with schizophrenia to:
 - Clarify thinking and perceptions of reality
 - Reduce hallucinations and delusions
- Drug treatment must be consistent to be effective. Inconsistent dosage may aggravate existing symptoms or create new ones.

Prognosis

Good prognosis

- Old age of onset
- Female
- Married
- No family history
- Good premorbid personality
- High IQ
- Precipitants
- Positive symptoms
- Treatment compliance
- Good support
- Acute onset
- Presence of mood component

Poor prognosis

- Young age of onset
- Male
- Unmarried
- Family history
- Personality problems
- Low IQ
- No obvious precipitants
- Negative symptoms
- Poor treatment compliance
- Low support
- Insidious onset
- No mood component

Summary of Schizophrenia

- Schizophrenia – Spectrum of dysfunctions
 - Affecting cognitive, emotional, and behavioral domains
 - Positive, negative, and disorganized symptom clusters
- DSM-IV and DSM V-TR
 - Five subtypes of schizophrenia/ no more exist!
 - Includes other disorders with psychotic features
- Several Bio-Psycho-Social variables are involved
- Successful treatment rarely includes complete recovery



Other Psychotic Disorders

- Schizophreniform Disorder
 - Schizophrenic symptoms for a few months (less than 6 months)
 - Associated with good premorbid functioning
 - Most resume normal lives
 - The same treatments recommended for schizophrenia may also be utilized here

- Brief Psychotic Disorder

- One or more positive symptoms of schizophrenia
- Usually precipitated by extreme stress or trauma
- experience a psychosis, while lasting at least a day, undergoes a full, complete and spontaneous remission within one month
- Tends to remit on its owns

- Delusional Disorder

- Delusions that are contrary to reality
- Lack other positive and negative symptoms
- Types of delusions include
 - Erotomantic, Grandiose, Jealous, Persecutory, Somatic
- appears to pursue a chronic, waxing and waning course
- Patients with paranoia rarely seek treatment with a psychiatrist on their own initiative
- Better prognosis than schizophrenia

- Shared Psychotic Disorder (**Folie à Deux**)

- Delusions from one person manifest in another person
- The most common relationships are among parents and children, spouses, and siblings
- Separation from the dominant person and immersion into normal social interaction

- Schizoaffective Disorder

- Symptoms of schizophrenia and a mood disorder
- There must be a period of at least two weeks of psychosis without mood disorder
- These symptoms cannot be due to medication, substance use or another medical condition
- Both disorders are independent of one another
- Such persons do not tend to get better on their own

- The mainstay of current treatment is antipsychotic medication combined with mood stabilizer medication or antidepressant medication, or both
- Psychotherapy vocational therapy and social/psychiatric rehabilitation are also important for recovery
- In cases where there is risk to self and others, brief involuntary hospitalization may be necessary
- long-term outcome of patients is not as good as that for patients with a mood disorder, yet not as grave as that for patients with schizophrenia

- Schizotypal disorder

- Characterized by eccentric behavior and by deviations of thinking and affectivity, which are similar to that occurring in schizophrenia, but without psychotic features and expressed symptoms of schizophrenia of any type
- May reflect a less severe form of schizophrenia

- Postpartum Psychosis (puerperal psychosis)
 - Rare disorder, occurring in perhaps less than 1 or 2 per 1000 deliveries
 - It is more common in primiparous than multiparous women
- Many of these patients never experience another psychotic illness unless they again become pregnant
- Symptoms generally appear abruptly within about 3 days to several weeks after delivery

- Hospitalization is generally indicated
- Severe overactivity and delusions may require rapid tranquilization by neuroleptic (antipsychotic) drugs
- Electro-convulsive (electroshock) treatment is highly effective
- Mood stabilizing drugs such as lithium are also useful in treatment and possibly the prevention of episodes in women at high risk

